A Review of Modern Approaches to Association Rule Mining in E-commerce Environments

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***Abstract*—The rapid evolution of digital commerce has resulted in the generation of massive transactional, behavioral, and navigational datasets, each reflecting different dimensions of user interaction within online platforms. Extracting meaningful, actionable insights from this data landscape requires robust data mining techniques capable of handling large-scale datasets, uncertainty in user behavior, heterogeneous data formats, and real-time processing demands. Among these techniques, Association Rule Mining (ARM) remains one of the most widely applied and adaptable analytical frameworks in e-commerce.**

The focus of this report is on modern methodologies of ARM, comparing and illustrating their application in a wide range of e-commerce-related environments. It offers an overview of relevant literature, from classical ARM algorithms like Apriori, FP-Growth, and ECLAT, which form the basis for frequent itemset mining; to enriched and domain-specific extensions like profit-based rule mining, clustering-assisted ARM, and fuzzy-set-driven rule extraction that address shortcomings in interpretability, utility, and uncertainty; and large-scale frameworks such as MapReduce-based sequential ARM combined with deep learning classifiers to assist with distributed environments and more complex behavioral modeling.

The examined studies collectively demonstrate the breadth of ARM applications in e-commerce, including personalized recommendation systems, retail purchasing pattern discovery, customer relationship modeling, and clickstream-based usability evaluation. These applications highlight ARM's ability to reveal hidden correlations among products, identify meaningful user navigation paths, detect high-value customer segments, and support data-driven decision-making for marketing, interface design, and user engagement strategies.

Among all the approaches reviewed, ARM stands out as highly versatile, and it remains key to understanding behavioral trends and optimizing operational workflows in modern e-commerce. This report points out relative strengths and weaknesses of algorithms and identifies emerging opportunities for improvement related to scalability, interpretability, and integration with machine-learning-based predictive models for future research in e-commerce.

***Keywords*—Association Rule Mining, E-Commerce Data Mining, FP-Growth, Apriori Algorithm, Fuzzy ARM, MapReduce, Sequential Pattern Mining, Clickstream Analysis, Recommendation Systems.**

I. INTRODUCTION

The rapid expansion of e-commerce platforms has led to an unprecedented surge in the volume, velocity, and variety of data generated from customer interactions. Every search query, product click, shopping cart update, and completed transaction contributes to a continuously evolving digital footprint. Extracting actionable knowledge from this data has become essential for supporting business intelligence, personalizing user experiences, optimizing product catalogs, and improving website usability. As a result, data mining has emerged as a foundational analytical tool in the e-commerce domain, enabling organizations to uncover patterns that are not readily observable through conventional statistical techniques.

Among the wide range of data mining methods, Association Rule Mining (ARM) has gained particular relevance for its ability to reveal hidden relationships among items, behaviors, and user actions. Classical ARM algorithms such as Apriori 1, FP-Growth 2 and ECLAT 3 were originally designed to identify frequent itemsets in retail transaction databases. These methods allow analysts to discover rules of the form A → B, indicating that customers who purchase or view item A are likely to purchase or view item B. Such rules support applications in cross-selling, product bundling, inventory planning, and recommendation systems. However, as e-commerce systems scale, traditional ARM approaches face increasing challenges related to computational complexity, memory limitations, and inability to incorporate uncertainty or domain-specific constraints.

To address these limitations, researchers have proposed several enhanced ARM frameworks tailored for modern e-commerce environments. One line of research extends classical algorithms using clustering techniques 4 or profit-aware rule mining , allowing recommendation systems

to consider not only item frequency but also item utility and customer segmentation. Another study investigates ARM within retail transaction environments, providing comparative insights on rule quality, algorithmic performance, and practical applicability. These methods highlight the adaptability of ARM when integrated with business-driven objectives. Other studies expand ARM's capabilities by incorporating fuzzy set theory, enabling rule extraction in scenarios where user preferences, product similarities, or behavioral indicators are not strictly binary. Fuzzy ARM frameworks address the ambiguity and uncertainty inherent in user-driven e-commerce platforms, where behaviors often fall into degrees rather than discrete categories. Similarly, MapReduce-based sequential ARM methods leverage distributed computing to scale rule mining to extremely large datasets. The integration of deep learning classifiers further enhances the interpretability and predictive power of the discovered rules, making these approaches suitable for real-time or near–real-time decision-making.

In addition to methodological advancements, ARM has been applied to diverse real-world e-commerce problems. One study investigates customer purchase behavior using over half a million transactions from a UK online retail dataset, applying Apriori, FP-Growth, and ECLAT to identify purchasing patterns and validate them using machine-learning models. Another study focuses on clickstream-based website usability analysis, converting thousands of user sessions into navigational sequences to discover frequent page transitions and identify drop-off points. These applications demonstrate how ARM supports both operational insights, such as demand forecasting and product affinity analysis, and user-experience improvements such as navigation optimization and checkout-flow refinement.

Despite differences in datasets, algorithms, and objectives, the reviewed papers share a common theme: ARM remains one of the most versatile and valuable tools for deriving meaningful insights from e-commerce data. Whether the goal is to improve recommendation accuracy, enhance customer engagement, optimize marketing strategies, or evaluate user interface effectiveness, ARM provides a structured analytical approach for uncovering non-trivial associations hidden within large-scale datasets.

This report presents a comparative examination of these ARM techniques and applications. By analyzing classical, fuzzy, utility-based, clustering-assisted, and distributed ARM frameworks alongside practical e-commerce case studies, we outline key contributions, limitations, and potential areas for future research. The goal is to highlight how evolving data mining methods continue to support both strategic and operational decision-making across the e-commerce ecosystem.

II. RELATED WORK

Paper 1: MapReduce Framework Based Sequential Association Rule Mining with Deep Learning Enabled Classification in Retail Scenario

This study addresses performance degradation, high resource consumption, and high computational overhead for traditional methods like Apriori when it comes to mining sequential association rules from large retail datasets and finding patterns in consumer purchasing behavior. The authors propose a hybrid approach that combines the Tasmanian Water Devil Optimization (TWDO) algorithm with Sequential Association Mining (SAM), implemented over the MapReduce framework. Data from retail transactions in the dataset is subdivided into several data blocks and is fed to the Mapper phase, where SAM is carried out, and optimal sequential association rules are filtered using the TWDO algorithm. In the reducer phase, the optimally selected sequential rules from the Mapper are combined to generate integrated 2-length and 3-length rules. These rules are then transformed into a rule matrix and passed into a Deep Convolutional Neural Network (DCNN) to classify users.

This approach was evaluated using data from the Frequent Itemset Mining Dataset Repository, using the number of rules mined, execution time, recall, and precision as metrics, against existing approaches such as the MR-Apriori mining algorithm, ARM, SHFIM, and CAN-FP growth. This approach outperformed its competition, producing the highest number of meaningful rules (183), the lowest processing time (7.70 seconds), strong precision (0.88), and recall (0.88). This shows that the proposed approach is able to produce better quality rules while being computationally efficient and accurate. Limitations of this approach could be that there are heavy computational requirements for TWDO, and there is a possibility for information loss when transforming rules into a rule matrix for the DCNN. The approach was also only tested using one dataset, and thus, generalizability is unclear.

Paper 2: Integrating Fuzzy Set Theory with Association Rule Mining for Integrating Fuzzy Set Theory with Association Rule Mining for Advanced E-Commerce Recommendations

This paper highlights the limitations of traditional Association Rule Mining (ARM), where it is unable to model uncertainty or represent linguistic attributes like price and quantity due to its Boolean nature. The author's approach to this is Fuzzy Association Rule Mining (FARM), which first fuzzifies price and quantity attributes. Fuzzification transforms numerical data into fuzzy values using predefined

membership functions. For example, features like product quantity or price can be expressed as fuzzy categories such as "Low," "Medium," or "High". The Apriori algorithm is then run on the fuzzified data for ARM.

FARM was carried out on three different datasets: the UCI Online Retail II Dataset, the Market Basket Analysis dataset, and the Movielens dataset, for thorough testing across various customer behaviour scenarios. It was evaluated using metrics such as support, confidence, and lift, and statistical tests such as the Chi-square and P-value tests. FARM was able to discover more refined rules, such as strong correlations between "Low Quantity" and "Cheap" products, supported by the metrics mentioned above. It achieved extremely strong Chi-square values and extremely low P-values, which indicate that the associations identified were highly statistically significant and unlikely to have occurred by random chance. This shows that the model is able to capture subtle and meaningful buying behavior patterns missed by traditional ARM. The strong results across three datasets from different domains also suggest a strong degree of generalizability. A limitation of this approach is that fuzzification increases computational cost and requires domain knowledge to define its membership functions. The paper also does not demonstrate how FARM would perform on continuously updating data streams, and thus, its applicability to real-time, large-scale e-commerce environments is still uncertain.

Paper 3: E-commerce Recommendation System using Association Rule Mining and Clustering

This paper by Vishal Parikh and Parth Shah from Nirma University, addresses the concerns of ARM-based recommender systems in real-time e-commerce applications. The initial issue was that running association mining rules on large transaction datasets would be too slow for real-time recommendations. To handle this, the authors split the data into three tables: Customers, Products, Transactions and define three recommendation modes:

1. Profile based (using gender, location and age group)

2. Social based (using a social network of friends who also shop on the site and recommend items friends have bought)

3. Cart based (using the items already in customers cart to recommend new ones)

They then incorporate hierarchical clustering, which all three modes rely on; it works by the data first getting split up by gender, then by age group, then by location, and finally by product. This method produces relatively small clusters which means that applying ARM on these clusters is much quicker and more efficient.

Experimentally, it took 6121ms to generate recommendations on average without clustering; and only about 98ms for a cluster with a maximum size of 100. And to create a cluster of maximum size 100 items, it took 1174ms on average. Hence, a total of 1272ms on average to cluster and recommend items to a specific user in contrast to 6121ms without clustering. This drop in timing shows that this method is reliable and could be used to speed up the process of ARM. In conclusion, this paper explores how incorporating basic pre-processing and combining hierarchical clustering with ARM-based recommender systems makes ARM feasible for near real-time use in e-commerce settings.

Here Table 1 and Fig.2 shows time to create cluster with respect to maximum size of the cluster.

Maximum size of cluster	Time to create cluster(in second)
100	1174
200	711
500	415
1000	298

Table 1.Maximum cluster size to Time for create cluster

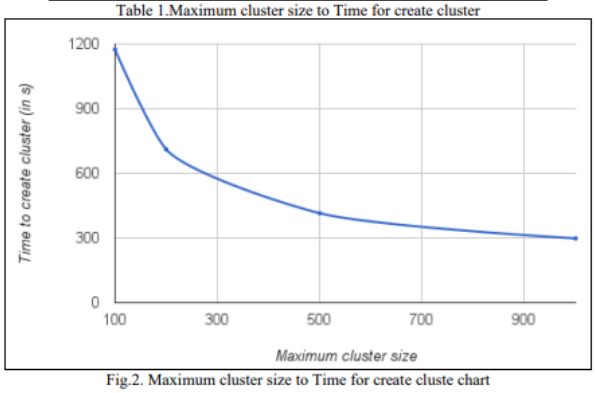

Fig.2. Maximum cluster size to Time for create cluste chart

Table 2 and Fig.3 shows time to generate recommendation with respect to maximum size of the cluster.

Maximum size of cluster	Time for generate recommendation(in milliseconds)
100	98
200	268
500	842
1000	1468
10000(without clustering)	6121

Table 2.Maximum cluster size to Time for generate recomandation

Paper 4: Association rule mining algorithm implementation for e-commerce in the retail sector

This study by Namatullah Wahidi and Rita Ismailova focuses on the application of Association Rule Mining to real retail e-commerce data from Kyrgyzstan. The authors analyze a data set from 2020 containing 577,730 invoices, around 7,652 customers, and 242 products from a department

store in Bishkek, which covered 8 municipalities and nine cities, with Bishkek accounting for 64% of total sales. By cleaning and organizing the data to concentrate on more popular items, they were able to reduce the data set to only 358,192 invoices. This reduction makes the data more manageable and processable which will later show why pre-processing data is of mass importance in the subject of ARM-based recommender systems.

After obtaining the clean dataset, they apply Apriori algorithm in python and further process the data with a minimum support of 40% and a minimum confidence of 80% to ensure they only obtain interesting associations in the form A → B (A implies B), this process yielded 118 strong association rules that captured frequent co-purchasing patterns.

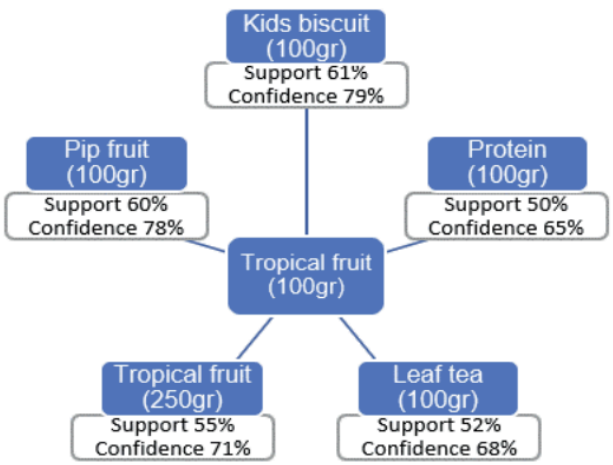

Figure 5: Relationship between products with highest supports

These rules are then interpreted as actionable insights for cross selling, targeted promotions and shelf layout optimization. Although this specific study is about a physical retail store, the principles used to find recommendations using Association Rule Mining can be used and applied to any e-commerce applications with large transaction history, demonstrating how classic Apriori and data mining pre-processing principles merged with Association Rule Mining based recommendation yield concrete business strategies in the e-commerce world from large-scale transaction datasets.

Paper 5: E-Commerce Website Usability Analysis Using the Association Rule Mining and Machine Learning Algorithm

This study investigates purchasing behavior using the publicly available UK Online Retail dataset, which contains over 500,000 transactions. The authors apply classical ARM algorithms: Apriori, FP-Growth, and ECLAT, to identify item co-occurrence patterns valuable for cross-selling and recommendation tasks.

The Apriori algorithm begins by generating candidate itemsets and filtering them using minimum support thresholds. Its fundamental operation is based on:

$$\text{Support } (X \rightarrow Y) = \frac{\sigma(X \cup Y)}{N}$$

where $\sigma(X \cup Y)$ is the number of transactions containing both X and Y, and N is the total number of transactions. FP-Growth improves efficiency by compressing transactions into an FP-tree, avoiding candidate generation. ECLAT relies on vertical transactions and intersection-based support counting.

Key Contributions:

A. Frequent itemset extraction:

Strong rules were obtained with confidence values around 0.55–0.70 and lift values above 2, indicating meaningful product affinities.

B. Algorithmic comparison:

FP-Growth demonstrated significantly faster runtimes -processing the dataset in under 2 seconds versus over 9 seconds for Apriori.

C. Predictive validation:

The extracted rules were validated using Decision Tree and Random Forest classifiers; Random Forest achieved approximately 86% classification accuracy, confirming the usefulness of the mined associations.

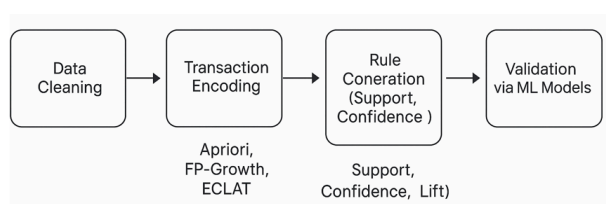

Figure 1. ARM Pipeline for customer purchase behaviour.

The pipeline outlines the process used to mine association rules from retail transaction data. Raw purchase logs undergo cleaning and are encoded into transaction format before being processed by Apriori, FP-Growth, or ECLAT. Frequent itemsets that satisfy minimum support thresholds are used to generate association rules based on support, confidence, and lift. The resulting rules are evaluated using machine-learning classifiers to determine their predictive strength and their ability to represent real purchasing behavior.

Paper 6: Mining Association Rules for Optimizing Customer Purchase Behavior in E-Commerce Transactions

This paper focuses on understanding e-commerce clickstream data-page-level navigation logs reflecting how users move across a website. After sessionizing raw logs, each user path is modeled as a sequence, and ARM is used to identify frequent transitions and potential usability bottlenecks.

User navigation is represented as sequences such as:

$$U_i = \{P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow \ldots\}$$

Where P_k denotes a visited webpage. Associations among pages are evaluated using traditional ARM metrics:

$$\text{Confidence } (X \rightarrow Y) = \frac{Support(X \cup Y)}{Support(X)}$$

Key Contributions:

A. Frequent navigation patterns:

One of the most common user flows: Home → Search → Product, appeared in roughly 30% of sessions with high confidence.

B. Lift analysis for path effectiveness:

The transition Product → Add to Cart exhibited a lift value near 1.8, indicating a strong conversion-oriented pattern.

C. Usability diagnostics:

Certain product-detail pages showed drop-off rates approaching 40%, revealing sections of the interface contributing to user abandonment.

Figure 2. Clickstream-Based Association Rule Mining Workflow

Each session is converted into a sequence of visited pages, enabling the discovery of high-confidence transitions and drop-off points within the website. The rules produced help reveal common user journeys and highlight interface areas that may require usability improvements.

Home → Search → Product → Cart → Checkout

Paper 7: Profit-Based Association Rule Mining (PARMA)

In the domain of e-commerce recommendation services, standard ARM algorithms tend to run into many significant limitations when applied to real-world profitability metrics. Dai and Zeng [7] addressed a critical inefficiency in traditional algorithms like Apriori, which is the failure to distinguish between high-frequency versus high-profit items. Standard algorithms rely on a single minimum support threshold, which assumes that all items are of equal nature and interest. This assumption leads to the "rare item problem," where profitable but infrequent items (e.g. a MacBook) are pruned early because they fail to meet global support thresholds. Attempting to correct this issue by lowering the support threshold to account for these rare items ends up creating a secondary issue: an explosion of meaningless rules involving frequent, low-margin items.

To address this issue, the authors proposed the Profit-support Association Rule Mining Algorithm (PARMA). This algorithm changes the mining criteria from being purely focused on sales volume to maximizing profit. The core aspect of PARMA is the introduction of a "Unique Profit" criterion, which assigns a specific minimum support to each item based on its unit profit, rather than a global percentage. The algorithm works by calculating a "Minimum Profit Support" (MPS) for every item, defined mathematically as the Appointed Profit divided by the Unit Profit. This helps make sure that items contributing equally to the total profit are treated with equal weight, regardless of their raw transaction frequency. The method also uses a pre-processing step to filter 1-itemsets based on minimum total profit, which removes low-value items earlier in the process before the more computationally expensive N-itemset mining begins.

The proposed solution was validated using the IBM synthetic data generator to create a controlled experimental environment consisting of 10,000 transactions and 1,000 items. To simulate a more realistic retail environment, profits were assigned using a simplified normal distribution: 80% of items were assigned medium profit ($1-5), 10% high profit ($5-10), and 10% low profit ($0.1-1). For this experiment, PARMA was compared against the MMSapriori (Multiple Minimum Support Apriori) algorithm. The results of the experiment showed that PARMA significantly improved efficiency by generating fewer large itemsets, filtering out 5,070 versus 5,140 at the 0.1% support level, while discovering a higher number of candidate itemsets when compared to MMSapriori (345,450 versus 326,520 at the 0.1% support level). These results confirmed that integrating profit constraints into the support calculation helps efficiently prune the search space, while still keeping rules that are economically significant.

Paper 8: Comparative Analysis of Candidate Generation vs. Pattern Growth

As e-commerce datasets grow in dimensionality and sparsity, the issue of selecting an appropriate algorithm for rule discovery becomes critical for both performance and scalability. Hunyadi et al. [8] identified a significant gap in the existing literature: while Apriori and FP-Growth are widely studied, there is a lack of statistically rigorous, tool-based comparisons using real-world e-commerce data, rather than synthetic benchmarks. Most of the previous studies were largely focused on theoretical concepts or small-scale testing. As a result, they often overlooked the practical reality, specifically, how adjusting support thresholds impacts performance in an actual operational environment.

To mitigate these issues, the study used a real-world dataset from an e-commerce platform spanning a six-month period, initially consisting of approximately 27,000 transactions. To ensure data quality, the authors applied rigorous preprocessing steps, such as: items appearing in fewer than 10 transactions were removed to reduce dimensionality, and transactions containing fewer than two items were excluded. The final refined dataset contained 22,476 transactions and 1,215 unique items with a density of 0.35%, which is typical of sparse retail environments. The authors implemented both algorithms in RapidMiner and validated the results using IBM SPSS for advanced statistical testing. The experiment involved varying minimum support (0.10, 0.15, 0.20) and confidence (0.10 to 1.00 in 0.05 increments). To ensure statistical validity, the Kolmogorov-Smirnov test was first used to assess normality. This test determined that FP-Growth results followed a normal distribution, which allowed the use of ANOVA. However, Apriori results at higher support levels were non-normal, requiring the Kruskal-Wallis test.

A critical finding of this experiment was related to memory efficiency. At the lowest support threshold (0.10), Apriori consumed approximately 780 MB of memory, whereas FP-Growth required only 430 MB, yielding a substantial 45% reduction. This efficiency is due to FP-Growth's compact tree structure, which eliminates the need for candidate generation and repeated database scans. FP-growth was also found to generate many more rules than Apriori, even at the lowest support. Furthermore, the study established a strong positive linear correlation of approximately 0.97 between the rule counts of the two algorithms, suggesting that the output of one algorithm can serve as a reliable predictor of the other under similar parameter configurations. The authors concluded that FP-Growth is better for large datasets and real-time recommendation systems, while Apriori may be preferable for teaching or when using smaller datasets.

III. DISCUSSION

Reviewing these eight articles reveals a clear trend: Association Rule Mining (ARM) is moving beyond simple statistical counting, and evolving into a tool that understands complex business needs. The reviewed literature suggests that modern e-commerce strategies can no longer rely on "one-size-fits-all" algorithms. Instead, successful implementation requires navigating three critical trade-offs: balancing computational speed with hardware limits, prioritizing economic value over raw frequency, and integrating interpretable rules with powerful predictive models.

The most immediate trend identified across the studies is the struggle to manage the sheer volume of modern datasets. Foundational algorithms, while theoretically sound, often break down under the weight of real-world e-commerce traffic. The comparative analysis by Hunyadi et al [8] and Kumar et al. [5] provides clear evidence of this bottleneck. While Apriori might serve as an excellent teaching tool, its mechanism requires generating large numbers of candidate sets and repeatedly scanning the database to count them. In a live retail environment, this process consumes too much memory, and can cause system crashes. The transition to FP-Growth is presented in the literature as the logical solution to this problem. By compressing the entire transaction database into a compact tree structure (the FP-Tree), the algorithm avoids the need for repeated scans, and largely reduces memory consumption. However, the literature also reveals that better algorithms alone are not enough for "Big Data." As datasets grow into millions of transactions, even efficient trees become too large for a single computer. Here, the research presents two diverging architectural solutions:

1. Intelligent Segmentation (Vertical Scaling):

Parikh and Shah [3] demonstrated that by clustering users into demographic groups (like age or location) before running the mining algorithm, the system becomes significantly faster. This approach argues that the key to speed is reducing the size of the "haystack" before searching for the needle.

2. Distributed Computing (Horizontal Scaling):

In contrast, Matrouk et al. [1] utilized the MapReduce framework to distribute the workload across multiple nodes. This suggests that for massive retailers, the solution isn't just a different algorithm, but a hardware infrastructure that supports parallel processing.

A major blind spot in traditional mining is the assumption that if an item sells often, it must be important.

Dai and Zeng [7] prove this is often wrong. In retail, cheap items sell constantly, while expensive luxury items sell rarely. For example, the selling frequency of bread (low-cost item) versus a MacBook (high-cost item). Standard algorithms often filter out these "rare" high-profit items because they don't meet the frequency threshold. The introduction of Profit-Based ARM fixes this by focusing on utility rather than just popularity. Similarly, the research on Fuzzy ARM [2] addresses the disconnect between rigid computer logic and actual human reasoning. In a standard database, a price is just a number. But to a human, that price is either "cheap," "moderate," or "expensive", which are concepts that change based on context. Instead of relying on strict binary values, fuzzy logic is used to process these linguistic terms. As a result, the algorithm produces rules that are far more interpretable.

While most studies focused on what customers buy (transactional data), the work on clickstream analysis [6] shows the importance of analyzing how customers move (navigational data). Transactional ARM tells a store which products go together, but it cannot explain why a user left the site without buying. By treating page views as a sequence (Home → Search → Product), ARM can identify "drop-off" points in the user interface. This suggests that ARM has a dual role in modern e-commerce: a backend role for inventory and recommendations, and a frontend role for optimizing the user experience and website design.

Finally, the most significant advancement identified is that ARM is no longer working in isolation. The most sophisticated approaches [1],[5] treat ARM not as the final decision-maker, but as a filter. Deep Learning models are incredibly powerful, but they often function as "black boxes", giving an answer without explaining why. By using ARM to find the initial, explainable patterns and then feeding those into a Neural Network or Random Forest for the final prediction, researchers can achieve the best of both worlds. The ARM component provides transparency and filters out noise, while the ML component handles the complex, non-linear relationships that simpler rules might miss. This hybrid approach represents the future of the field, moving from static statistical reporting to dynamic, predictive artificial intelligence.

IV. CONCLUSION

In conclusion, association rule mining has evolved far beyond simple statistical counting. It has now become a comprehensive framework for understanding e-commerce dynamics. This review explored its wide-ranging applications, from analyzing purchase behavior to optimizing site navigation. We observed that while classical methods like Apriori and FP-Growth set the foundation, newer variations are addressing complex real-world limitations. For instance, PARMA prioritizes profitability over frequency, MapReduce solves the problem of scale, and Fuzzy ARM bridges the gap between raw data and human reasoning.

Although the field is still developing, the underlying goal of these ARM techniques remains consistent throughout all papers, which is to transform data into decision-making power. As these methods continue to mature, they offer a powerful pathway to increased customer satisfaction and sustained business growth.

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